

NUMPY: AVERAGE SALE PRICE OF HOUSES

AIM

To write a Python program using NumPy to find the average sale price of houses with more than four bedrooms.

ALGORITHM

1. Start.
2. Import the NumPy library.
3. Create a NumPy array named `house_data` where each row contains the number of bedrooms, square footage, and sale price, in that order.
4. Select rows where the number of bedrooms is greater than four.
5. Extract the sale-price column from the selected rows.
6. Calculate the average sale price using `np.mean()`.
7. Display the average sale price. If no house has more than four bedrooms, display a suitable message.
8. Stop.

PROGRAM (PYTHON)

```
import numpy as np

# Columns: bedrooms, square footage, sale price
house_data = np.array([
    [3, 1500, 250000],
    [5, 2200, 400000],
    [4, 1800, 320000],
    [6, 2800, 500000],
    [5, 2400, 450000]
])

# Select houses with more than 4 bedrooms
selected_houses = house_data[house_data[:, 0] > 4]

if len(selected_houses) > 0:
    average_price = np.mean(selected_houses[:, 2])
    print("Average sale price:", average_price)
else:
    print("No houses have more than 4 bedrooms.")
```

INPUT

Sample `house_data` array (columns: bedrooms, square footage, sale price):

Bedrooms	Square Footage	Sale Price
3	1500	250000
5	2200	400000
4	1800	320000
6	2800	500000
5	2400	450000

OUTPUT

Average sale price: 450000.0

RESULT

The average sale price of houses with more than four bedrooms was calculated successfully. For the sample data, the average sale price is 450,000.